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EXPERIMENT 5

Title: Automating Application Deployment using GitHub and Jenkins

Aim: To automate the application build, testing, and deployment process using GitHub and Jenkins.

Objectives

- To create a GitHub repository.
- To create and maintain source code using GitHub.
- To install and configure Jenkins.
- To integrate Jenkins with GitHub.
- To create a Jenkins Pipeline.
- To automate Build, Test, and Deploy stages.
- To verify successful pipeline execution.

Requirements:

Hardware

- Computer/Laptop
- Internet connection

Software

- Windows OS
- Git
- Java JDK 21
- Jenkins
- GitHub
- Web browser

Procedure

Step 1: Create GitHub Repository

Create a GitHub repository named:
jenkins-demo
Initialize the repository with a README file.

Step 2: Create index.html

Create a file named:
index.html
Add the following code:

```
<!DOCTYPE html>
<html>
```

```
<head>
   <title>Jenkins Demo</title>
</head>
<body>
   <h1>Hello Jenkins!</h1>
</body>
</html>
```

Commit the file to the main branch.

Step 3: Install Java

Install Java JDK 21.

Verify the installation using:

java -version

Java 21 should be displayed.

Step 4: Install Jenkins

Install Jenkins on Windows.

During installation, select the Java 21 installation:

C:\Program Files\Eclipse Adoptium\jdk-21.0.12.101-hotspot\

Run Jenkins as a Windows service.

Use port:

8080

Step 5: Open Jenkins

Open a web browser and enter:

http://localhost:8080

Unlock Jenkins using the initial administrator password.

Step 6: Install Plugins

Select:

Install suggested plugins

Wait until the plugin installation is completed.

Step 7: Create Administrator Account

Create a Jenkins administrator account by providing:

Username
Password
Full Name
Email Address

Then open the Jenkins dashboard.

Step 8: Create a Pipeline Project

From the Jenkins dashboard, select:

New Item → jenkins-demo → Pipeline → OK

Step 9: Create Pipeline Script

Under Pipeline, select:

Pipeline script

Enter the following script:

```
pipeline {
   agent any

   stages {
      stage('Build') {
         steps {
            echo 'Building application...'
         }
      }

      stage('Test') {
         steps {
            echo 'Testing application...'
         }
      }

      stage('Deploy') {
         steps {
            echo 'Deploying application...'
         }
      }
   }
}
```

Click Save.

Step 10: Run the Pipeline

Click:

Build Now

Open the generated build number and select:

Console Output

Step 11: Create Jenkinsfile

In the GitHub repository, create a file named:

Jenkinsfile

Add:

```
pipeline {
   agent any

   stages {
      stage('Build') {
         steps {
            echo 'Building application...'
         }
      }

      stage('Test') {
         steps {
            echo 'Testing application...'
         }
      }

      stage('Deploy') {
         steps {
            echo 'Deploying application...'
         }
      }
   }
}
```

Commit the file.

The repository should contain:

README.md
index.html

Jenkinsfile

Step 12: Connect Jenkins with GitHub

Open:

Jenkins → jenkins-demo → Configure → Pipeline

Change Definition to:

Pipeline script from SCM

Select:

SCM: Git

Repository URL:

https://github.com/gunjanrupani/jenkins-demo.git

Branch:

*/main

Script Path:

Jenkinsfile

Click Save.

Step 13: Execute the GitHub Pipeline

Click:

Build Now

Jenkins obtains the Jenkinsfile from GitHub and checks out the latest source code.

The pipeline executes:

Build
↓
Test
↓
Deploy

Step 14: Verify Output

Open:

Build Number → Console Output

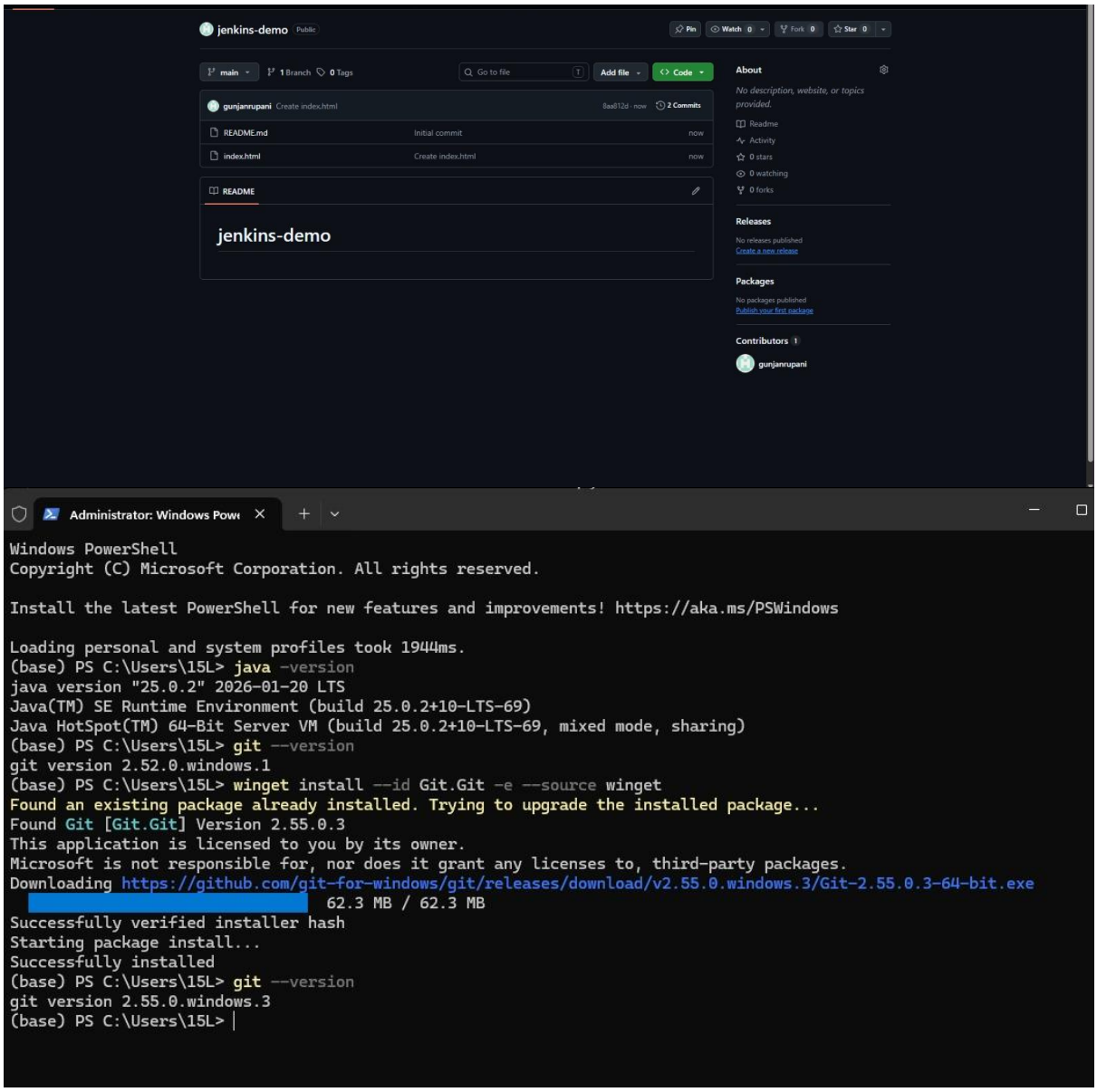
Step 15: Modify the Application

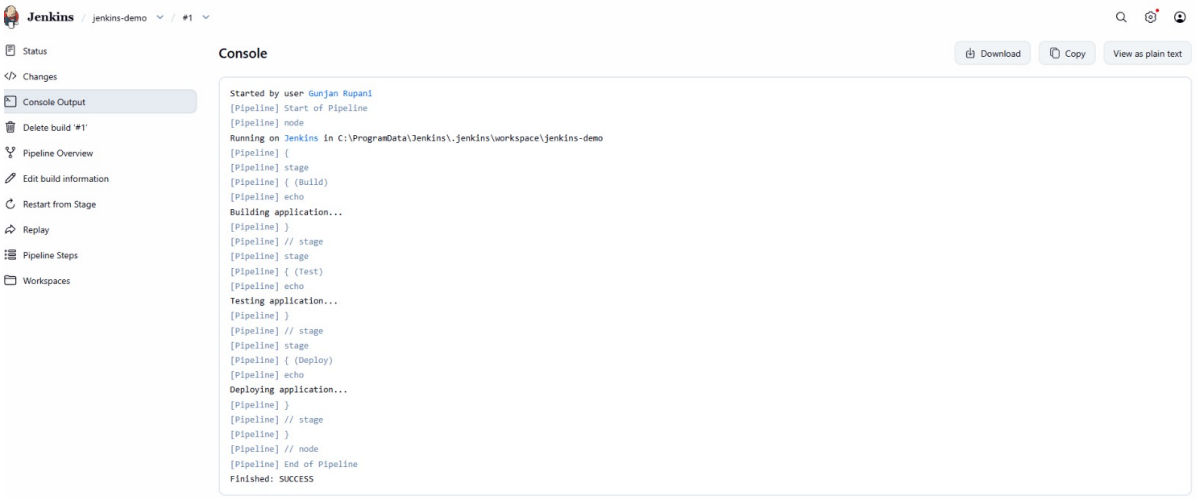
Update index.html in GitHub:

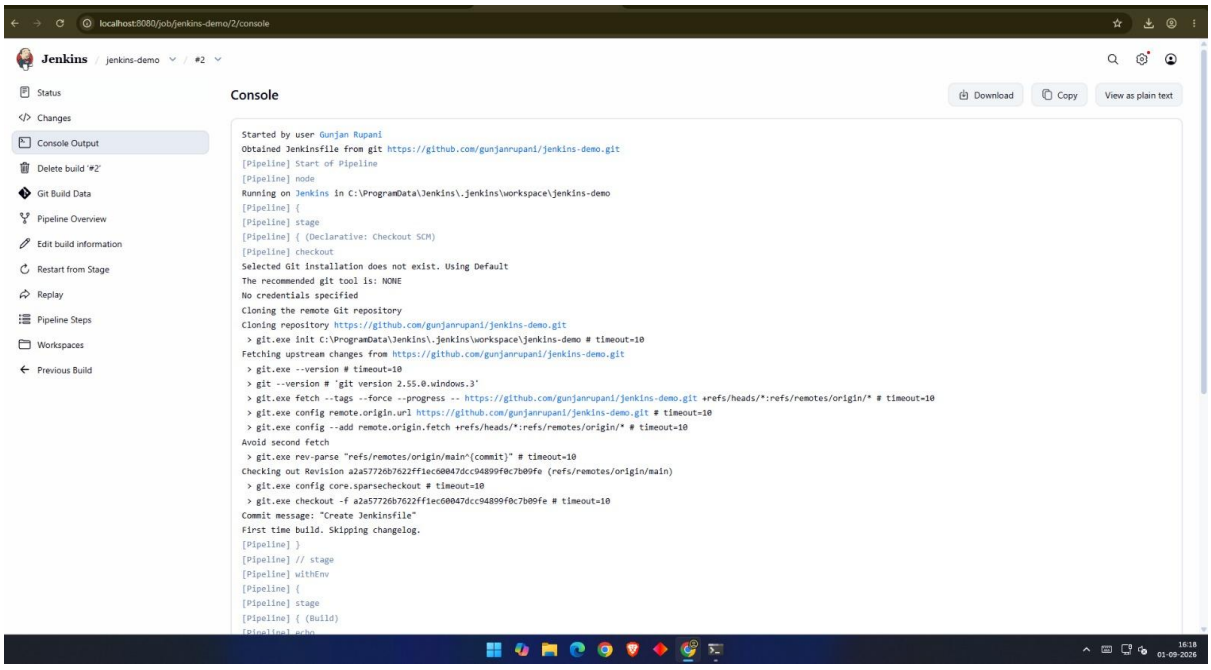
<h1>Hello Jenkins! Version 2</h1>

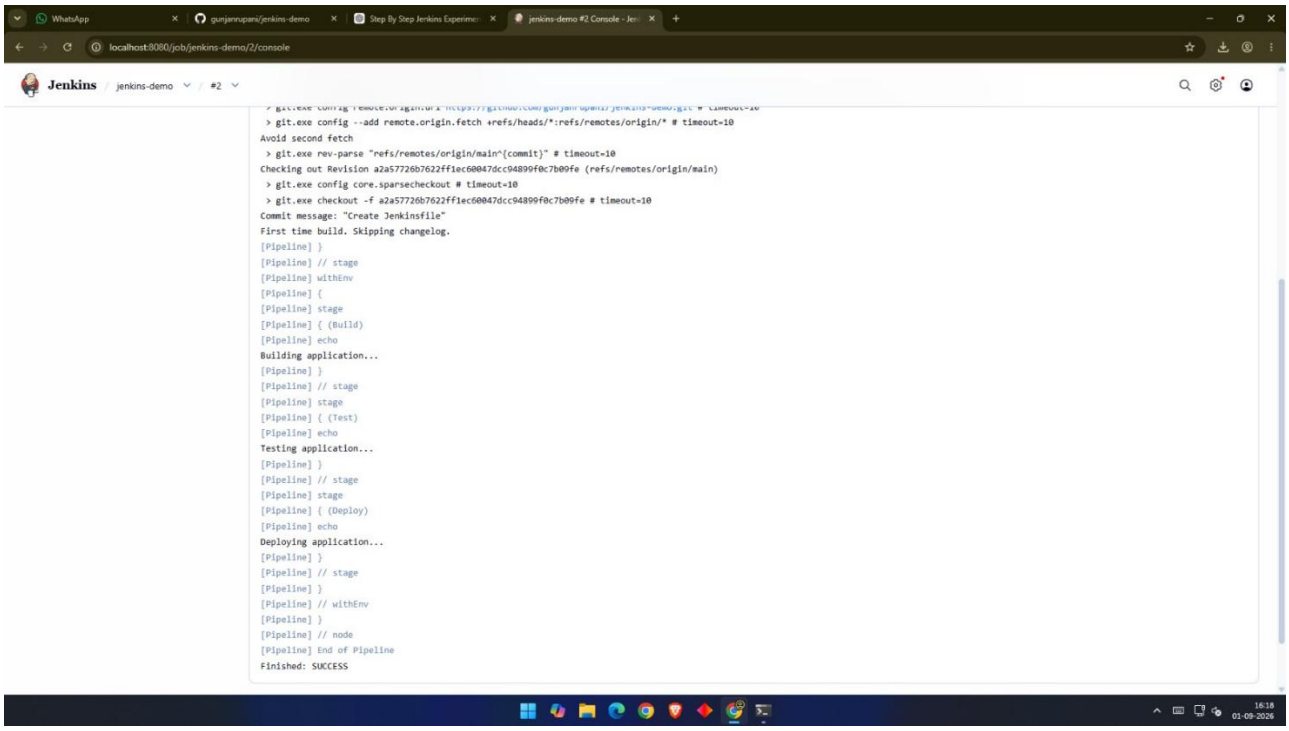
Commit the change.
Run Build Now again in Jenkins.
Jenkins retrieves the updated commit and executes the Pipeline.

Output:




Jenkins / jenkins-demo / #1
Status
Changes
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user Gunjan Rupani
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\jenkins-demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Testing application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.568.2


localhost:8080/job/jenkins-demo/2/console
Jenkins / jenkins-demo / #2
Status
Changes
Console Output
Delete build '#2'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user Gunjan Rupani
Obtained Jenkinsfile from git https://github.com/gunjanrupani1/jenkins-demo.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\jenkins-demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/gunjanrupani1/jenkins-demo.git
> git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\jenkins-demo # timeout=10
Fetching upstream changes from https://github.com/gunjanrupani1/jenkins-demo.git
> git.exe --version # timeout=10
> git --version # 'git version 2.55.0.windows.3'
> git.exe fetch --tags --force --progress -- https://github.com/gunjanrupani1/jenkins-demo.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe config remote.origin.url https://github.com/gunjanrupani1/jenkins-demo.git # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision a2a57726b7622ff1ec60047dcc94899f0c7b09fe (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f a2a57726b7622ff1ec60047dcc94899f0c7b09fe # timeout=10
Commit message: "Create Jenkinsfile"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
16:18
01-09-2026



Result

The GitHub repository was successfully integrated with Jenkins. Jenkins successfully retrieved the Jenkinsfile from GitHub and executed the Build, Test, and Deploy stages. The Pipeline completed successfully with:

Finished: SUCCESS

Conclusion

Thus, the application build, testing, and deployment process was successfully automated using GitHub and Jenkins Pipeline.